AMR GOMAA

Applied Machine Learning Research Deputy Head & PhD Candidate

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RESEARCH EXPERIENCE

Deputy Head and Researcher (Ph.D.) in Human-centered AI & Applied Machine Learning

German Research Center for Artificial Intelligence (DFKI)

March 2020 – March 2024 | Saarbrücken, Germany

- I am broadly interested in research areas related to the intersection of **machine learning** (language and vision) with **Human Computer Interaction**.
- My research topics include **Multimodal Gesture Recognition**, **Incremental Learning**, **Reinforcement Learning**, and **LLMs** for Automotive, Robotics, and Personalization.
- I have worked on multiple interdisciplinary industry-oriented projects with industrial partners such as **BMW**, **IAV (Volkswagen Subsidiary)**, and **Carl Zeiss**.

Project Manager, TeachTAM Project

German Research Center for Artificial Intelligence (DFKI)

March 2022 – March 2024 | Saarbrücken, Germany

- I created a project idea and established its desired goals, outcomes, milestones and acquired research funding from the German Federal Ministry of Education and Research (BMBF).
- I was responsible for leading a team of researchers in terms of coordination, organization, supervision and technical support to ensure project goals fulfillment in a timely manner.
- I coordinated and reported the results to project stakeholders (e.g. industry partners and the BMBF).

Junior Researcher

German Research Center for Artificial Intelligence (DFKI)

April 2019 – March 2020 | Saarbrücken, Germany

- Developed and Implemented human-machine interfaces for multimodal interaction with autonomous systems (vehicle, robot) including Hardware integration
- Training end-to-end machine learning (ML) systems for such interfaces including data collections and evaluations

Student Research Assistant

HCI Group, Saarland University

Oct. 2017 – April 2019 | Saarbrücken, Germany

- Worked on wearable devices including prototyping and novel circuit design including printed electronics, on-skin circuits and micro-controller programming creating tactile feedback with ultra thin electrons. This work was published in top HCI conferences such as UIST.

EDUCATION

Ph.D. in Computer Science

Deutsches Forschungszentrum für Künstliche Intelligenz and Saarland University

March 2020 – March 2024

- Focus on Incremental and Continual Learning approaches for Automotive and Robotics
- Supervisors: Dr. Michael Feld and Prof. Dr. Antonio Krüger

M.Sc. in Computer Science

Saarland University, Germany

Oct. 2017 – Feb. 2020

- GPA: 1.4/1.0
- Courses include: Machine Learning and Deep learning, Statistical Analysis, Human Computer Interaction, Image Processing and Computer vision, Embedded Systems and Embedded Security
- Supervisors: Dr. Michael Feld and Prof. Dr. Antonio Krüger

B.Sc. in Telecommunications and Electronics Engineering

Ain Shams University, Egypt

Aug. 2008 – Aug. 2013

- Grade: Very good with Honor (out of: pass, good, very good, excellent)
- Thesis title: Automatic Handover Optimization in 4G: LTE Self Optimizing Network (LTE-SON)

SKILLS

Natural Language Processing | LLMs | Machine Learning | Deep Learning | Generative Models | RNNs | LSTMs | CNNs | Transformers | scikit-learn | PyTorch | Computer Vision | AI | Multimodal Interaction | Python | Reinforcement Learning | Keras | R | TensorFlow | HCI | Project Management

SELECTED PUBLICATIONS

- Sahar Abdelnabi, Amr Gomaa, et al. **LLM-Deliberation: Evaluating LLMs with Interactive Multi-Agent Negotiation Games** @ ICLR '2024 LLMAgents Workshop.
- Amr Gomaa, et al. **Toward a Surgeon-in-the-Loop Ophthalmic Robotic Apprentice using Reinforcement and Imitation Learning**. In submission.
- Amr Gomaa, et al. **Unveiling the Role of Expert Guidance: A Comparative Analysis of User-centered Imitation Learning and Traditional Reinforcement Learning.** @ HITLAML '2023 Workshop. Best Paper Award.
- Amr Gomaa, et al. **Looking for a better fit? An Incremental Learning Multimodal Object Referencing Framework adapting to Individual Drivers** @ IUI '2024.
- Amr Gomaa, et al. **SynthoGestures: A Multi-Camera Framework for Generating Synthetic Dynamic Hand Gestures for Enhanced Vehicle Interaction.** @ IEEE IV 2024.
- Amr Gomaa, et al. **t's all about you: Personalized in-Vehicle Gesture Recognition with a Time-of-Flight Camera.** @ AutoUI '23.
- Amr Gomaa, et al. **Towards Adaptive User-centered Neuro-symbolic Learning for Multimodal Interaction with Autonomous Systems** @ ICML '2023 AI&HCI Workshop and @ICMI '2023 Blue Sky Paper.
- Amr Gomaa, et al. **What's on your mind? A Mental and Perceptual Load Estimation Framework towards Adaptive In-vehicle Interaction while Driving.** @ AutoUI '22.
- Amr Gomaa, et al. **ML-PersRef: A Machine Learning-based Personalized Multimodal Fusion Approach for Referencing Outside Objects From a Moving Vehicle.** @ ICMI '21.

INDUSTRY EXPERIENCE

Senior Radio Frequency (RF) Optimization Engineer

Vodafone Egypt

May 2016 – September 2017 Cairo, Egypt

- Responsible for the RF design, developments, optimization, and RF engineering services of Alexandria (the second largest city in Egypt and a fifth of Egypt's Entire Cellular Network). Also responsible for network design, development, roll out, integration and implementation of 2G/3G/4G wireless network systems.
- Worked with systems and IC designers to optimize power optimization for all RF circuitry in the cellular system. I was the project manager for several multi-functional teams targeted at optimizing RF dependencies to enhance Key Radio Features.

Vodafone Discover Graduate Program

Vodafone Egypt

March 2015 – April 2016 Cairo, Egypt

- Discover is a global accelerated development program that enables talented fresh graduates to develop and grow within Vodafone, and ultimately become Vodafone's future leaders through; customized job rotations, challenging business exposure, continuous development, and corporate networking.
- Successfully joined a highly selective program to jump-start my career and fill a senior engineer position in a record time of only one year where I have worked with two highly experienced teams (RF Planning/Optimization & RAN Optimization).

ACADEMIC SERVICES

Reviewer for AI & HCI conferences

2020 - Ongoing

AC (Associate Chair) at AutomotiveUI 2024 & 2023, PC (Program Committee) at HITLAML 2023, IMWUT 2023, CHI (Special Recognition for Outstanding Reviews) 2024 & 2023, IEEE VR 2023, NordiCHI 2022, AutomotiveUI (Special Recognition for Outstanding Reviews) 2023 & 2022 & 2021, CHI PLAY 2021, ICMI 2021, and IEEE AIVR 2021.

Student Volunteer

2019 - Ongoing

Assisted organizers in UPLINX Machine Learning Summer School 2019, CIXSchool 2022, and AutomotiveUI 2022.

SCHOLARSHIPS AND AWARDS

- Won Third Place for Blue Sky Papers at ICMI'23, 2023
- Won Best Paper award at the HITLAML Workshop 2023
- Awarded the German Federal Ministry of Education and Research **(BMBF) research grant (100,000€) -** Software Campus Program (2021), TeachTAM Project
- Awarded **Saarland Scholarship** for international students at Saarland University - DAAD STIBET III scholarship grant (2019)
- Successful Selection and Participation in various Hackathons and Summer Schools such as the Sixth Summer School on Computational Interaction (**CIXSchool 2022**), the First Inria-DFKI European Summer School on AI (**IDESSAI 2021**), and **Continental**'s **Industry 4.0 & AI Hackathon** in the Deep Learning Track (2019)
- Successful Completion of Coursera's **Google Data Analytics** Professional Certification by Google on Coursera (including 8 courses), Entire **Reinforcement Learning Specialization** (including 4 courses), Practical Reinforcement Learning course with Honors, and **Embedded Hardware and Operating Systems course**

LANGUAGES

English (Bilingual)	●●●●●
Colloquial Egyptian (Bilingual)	●●●●●
German (Good)	●●●○○

TEACHING EXPERIENCE

Student Theses Mentor and Co-Superviser

Saarland University

2020 - Ongoing   Saarbrücken, Germany

- Kiran Gani, Master Thesis Title: Personalisation of Modalities using Reinforcement Learning for Object Referencing
- Simon Engel, Master Thesis Title: Adaptive User Interface Approach for Efficient Transfer of Control in Conditionally Automated Vehicles.
- Michael Sargious, Bachelor Thesis Title: An Adaptive Incremental End-to-End ML Framework with Automated Feature Engineering.
- Hassan Kanso, Bachelor Thesis Title: Multi-person Multi-camera tracking for Industry 4.0.

Lecturer

Saarland University

2019 - Ongoing   Saarbrücken, Germany

Selected Classes: Speech-based Adaptation of Personalized User Interfaces Seminar, AI for HCI Seminar, Adaptive Human Machine Interfaces for Autonomous Systems Seminar, Hybrid Machine Learning Approaches and Applications Seminar, Automotive User Interfaces Seminar, and Machine Learning School, and Machine Learning School.

Tutor

Saarland University

2017 - 2019   Saarbrücken, Germany

Courses: Human Computer Interaction and Neural Networks: Implementation and Application.

REFERENCES

Prof. Dr. Antonio Krüger

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Deutsches Forschungszentrum für Künstliche Intelligenz (DFKI)

Dr. Michael Feld

@ michael.feld@dfki.de

Deutsches Forschungszentrum für Künstliche Intelligenz (DFKI)

Prof. Dr. Anna Maria Feit

@ feit@cs.uni-saarland.de

Computer Science Department, Saarland University

Dr. Mridula Singh

@ michael.feld@dfki.de

CISPA – Helmholtz Center for Information Security